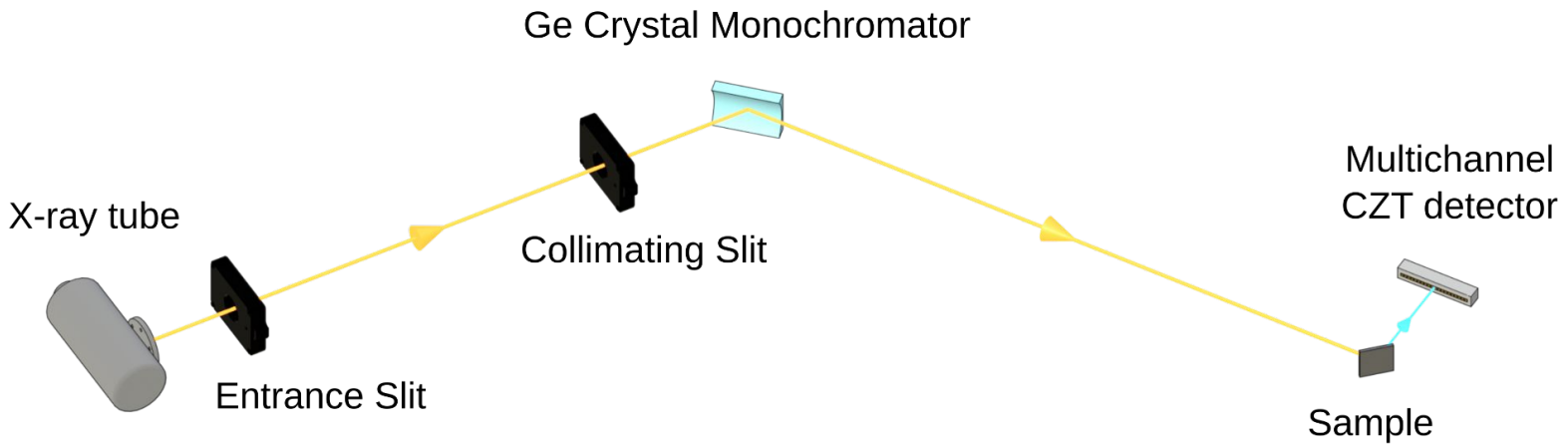
PRISM

Precision Room-temperature Instrumentation for high-energy X-ray absorption Spectroscopy Measurements

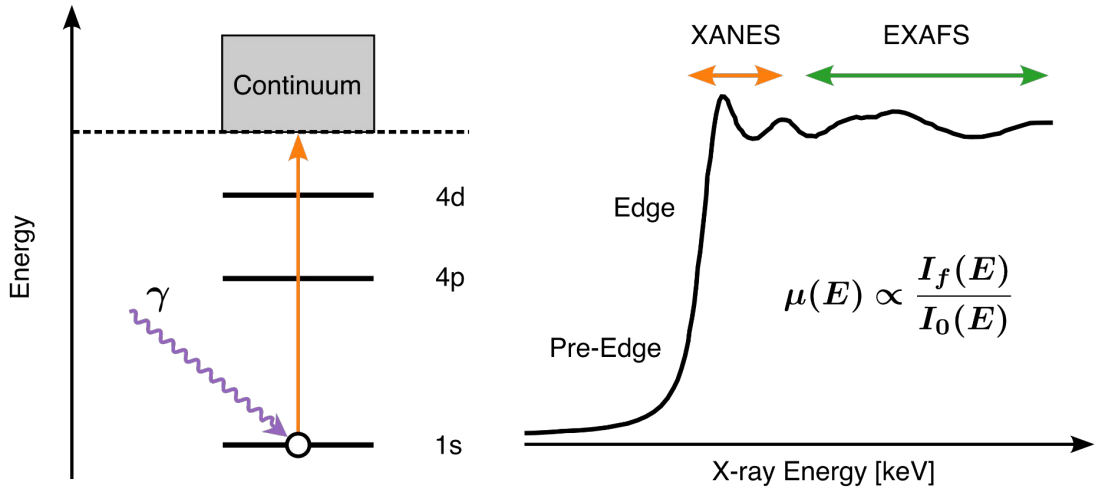
Responsabile di Progetto: Dr. Simone Manti, INFN - LNF
Referente di Progetto: Dr.ssa Catalina Curceanu, INFN - LNF

How can we perform laboratory XAS above 20 keV?



X-ray Absorption Spectroscopy (XAS): chemical state and local structure

XAS probes element-specific **Absorption Edges** to reveal **Chemical State** and the **Local Atomic Environment**.



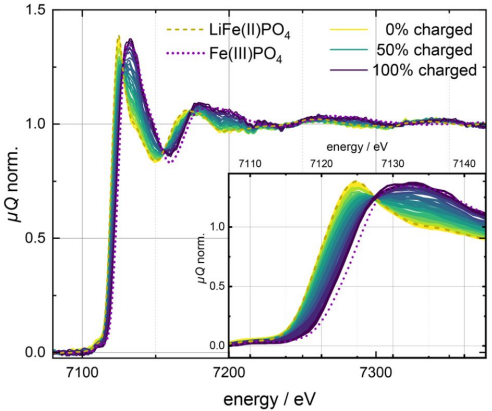
1. **Element selectivity**
   Core-level edges identify the absorbing element
2. **EXAFS**
   Neighbor distances, coordination and disorder
3. **Pre-edge**
   Electronic states and local symmetry
4. **XANES**
   Oxidation state and coordination geometry

State-of-the-Art XAS under operating conditions

Element-specific insight into **Electronic State** and **Local Atomic Structure**.

Battery cycling

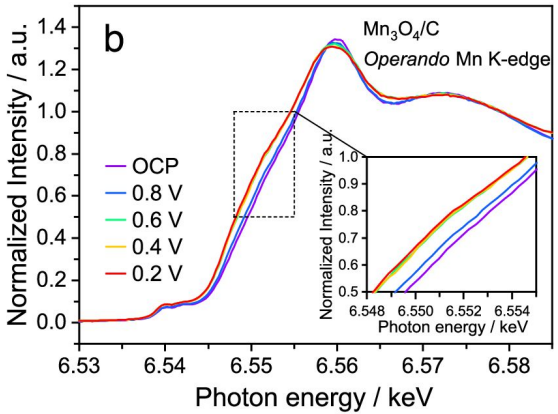
Lab XANES tracks $LiFePO_4$ conversion
Fe K-edge, 15 min per spectrum



Praetz et al., JAAS 40, 2447–2461 (2025), Fig. 2

Working catalysts

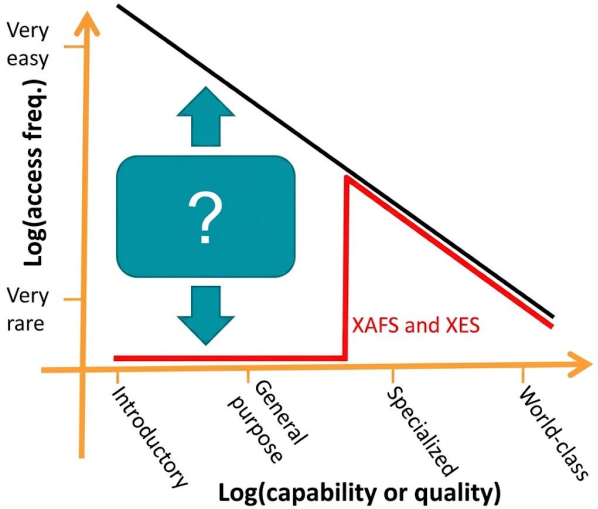
Operando XANES probes the Mn state
under fuel-cell operating conditions



Li et al., Nature Communications (2025), Fig. 3b

Why bring XAS into the laboratory?

Synchrotrons and **FEL** remain the reference for XAS performance.
Laboratory instruments with X-ray tubes offer routine access and flexible measurement times.



J. Seidler, et al. (2024)

- **Routine screening** and **repeated measurements**
- Long **In-Situ / Operando** campaigns
- Sample and protocol optimization for beamtime

Commercial laboratory XAS already exists!

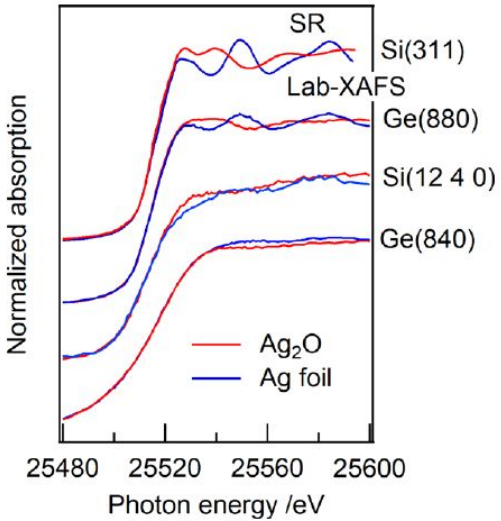


Sigray
≈ 1.5 M€



Laboratory XAS above 20 keV: opportunities and challenges

- K-edges of Mo, Pd, Ag and Sn are relevant to catalysis and energy materials.
- Closely spaced L-edges can restrict the accessible EXAFS range.



Feasibility demonstrated in transmission

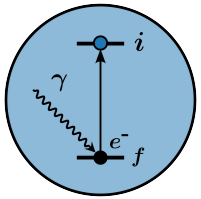
- Ge(880): ≈12 h for Ag K-edge XANES
- Ag K-edge EXAFS impractical at this count rate

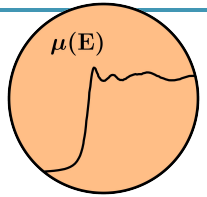
Spectral purity is also critical

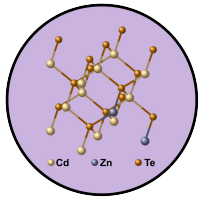
- Al filters suppress lower-order reflections and characteristic Ge K lines.

Challenge: quantitative fluorescence XAS with efficient detection and sufficient SNR!

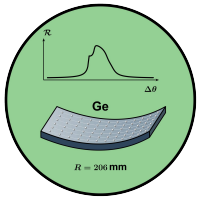
Yamamoto. "Selection of Monochromator Crystal for Pb L- and Ag K-Edge XANES Using a Laboratory-Based Spectrometer." *X-Ray Spectrometry* (2026).

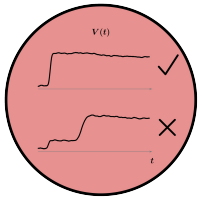






PRISM



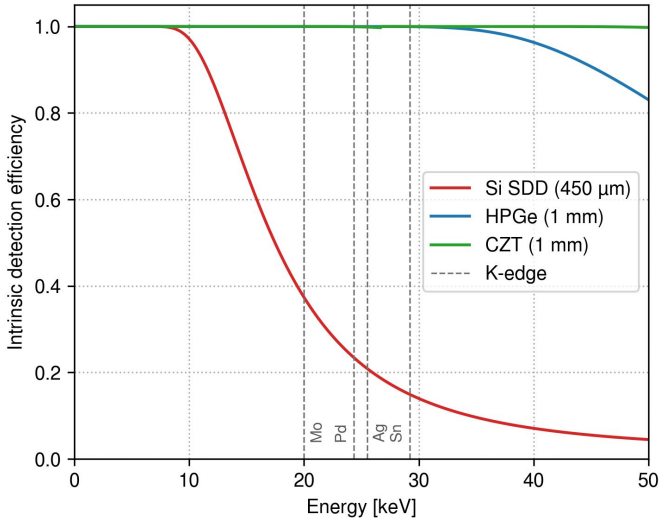


MAIN GOAL:

"To develop and validate a room-temperature laboratory platform for fluorescence-mode XAS above 20 keV, combining a high-resolution crystal monochromator with a custom multichannel CZT detector "

Why a multichannel CZT detector for PRISM?

Room-temperature detection of Ag fluorescence at 22.2 keV





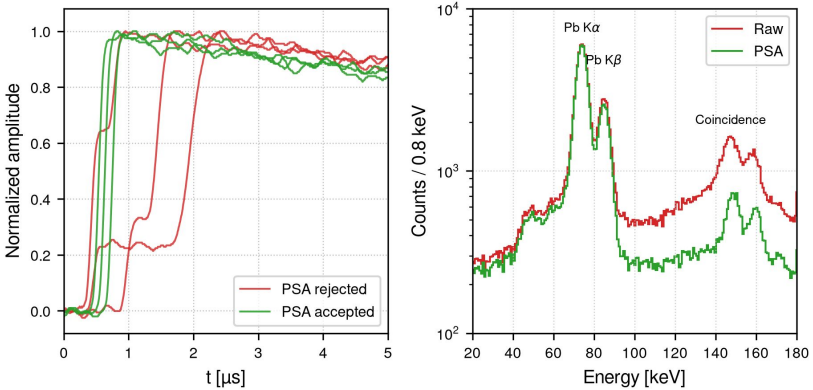
Illustrative pixelated sensor
PRISM baseline: 20-pixel linear array

- 2-mm-thick CZT
- 20 pixels, 500 µm pitch
- Independent low-noise channels distribute the rate
- Energy selection helps reject background

Calculated absorption: 450 µm Si, 1 mm HPGe, 1 mm CZT.

CZT development and qualification

Prior CZT results at DAΦNE BTF



Pulse selection preserves fluorescence peaks and suppresses coincidence structures.

Manti et al., Condens. Matter 11, 38 (2026).

Previous CZT measurements. PRISM targets remain to be validated.

Qualification targets at M15

Intrinsic efficiency >90% at 50 keV

Resolution ≤2% at Ag Kα (22.2 keV)

Dead time <10%
Pile-up contamination <5%

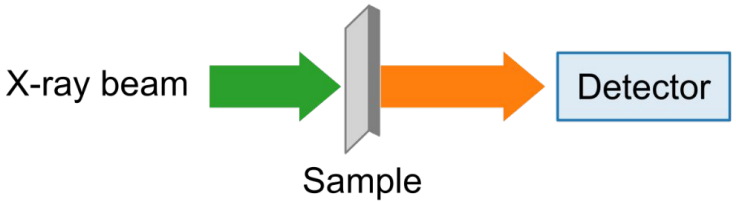
Under XAS operating conditions

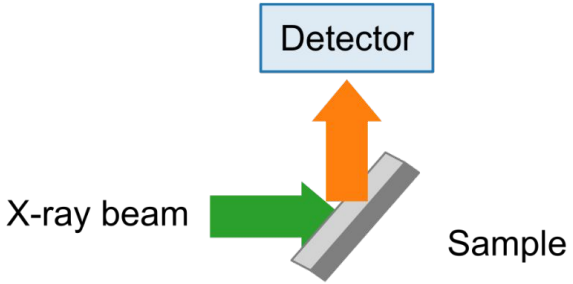
PRISM development
Pixel calibration and charge-sharing reconstruction
Waveform DAQ: 64 ch, 16 bit, 125 MS/s

Fluorescence detection mode for PRISM

Transmission Mode
requires a suitable, homogeneous sample thickness.

Fluorescence Mode
extends access to thick, dilute or constrained samples.

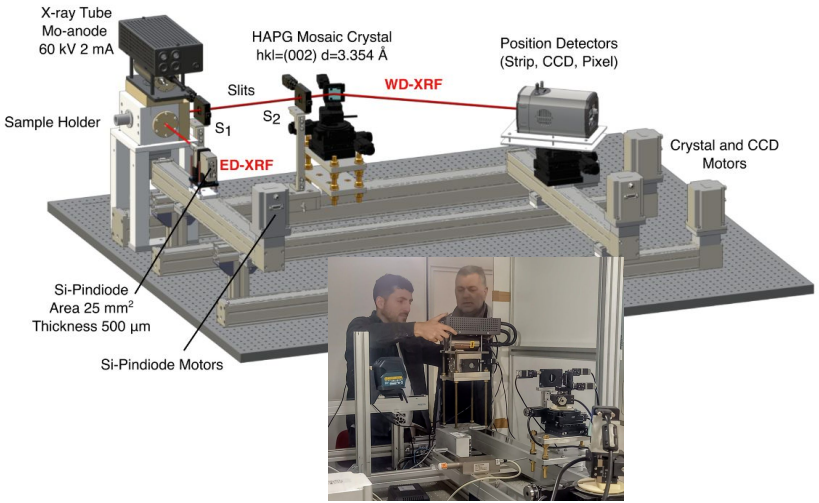


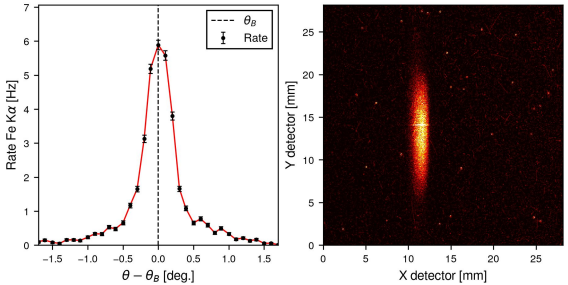


For **PRISM**, this enables measurements on realistic sample towards operating conditions

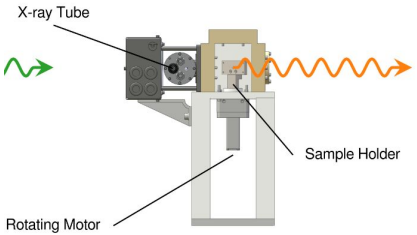
VOXES as the experimental basis for PRISM

Existing instrument and infrastructure





Angular scan and **focused beam** measured with the crystal optics.

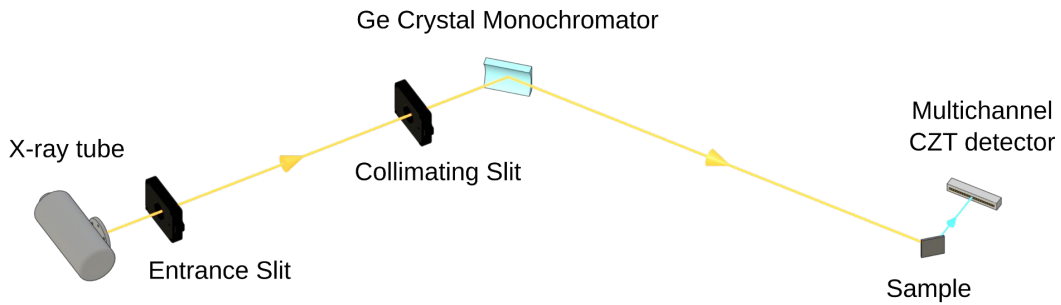


Recent upgrades enable **Absorption measurements**

Manti, Simone, et al. "Enhancing Performances of the VOXES Bragg Spectrometer for XES Investigations." Condensed Matter 9.1 (2024): 19.
Manti, Simone, et al.. "Recent Developments of the VOXES Von Hamos X-Ray Spectrometer for Laboratory XES and XAS Studies." Engineering Innovations 19 (2026): 23-29.

PRISM setup: selecting and scanning the incident energy

Ge(880) is the baseline candidate for the Ag K-edge at 25.5 keV.



Crystal rotation and motorized axes preserve the focused spot during the scan.

The same CZT measures If and I0 in separate motorized configurations.

SHADOW4 + GEANT4 optimize flux, bandwidth and background.

Ag K-edge XANES: the quantitative demonstration

Reference samples

Metallic Ag foil
Energy calibration and reproducibility

Two Ag(I) reference compounds
Different local coordination environments

Realistic sample
Test discrimination of chemical states

Acceptance targets

≤10 eV FWHM
Incident-beam bandwidth

≥20 edge-step SNR
Ag K-edge XANES

Reference-spectrum comparison
Energy scale and near-edge features

Extract Ag Kα, correct background and live time, then normalize to I0.

From instrument design to the Ag demonstration

WORK PACKAGE (WP)	PLANNED ACTIVITY (A)	2027				2028			
		1-3	4-6	7-9	10-12	13-15	16-18	19-21	22-24
WP1 - Requirements, simulations and design freeze	A1.1 - Requirements definition								
	A1.2 - SHADOW4 ray-tracing optimization								
	A1.3 - GEANT4 fluorescence simulations								
	A1.4 - Final geometry selection			MS1					
WP2 - Room-temperature multichannel CZT spectroscopy for hard X-ray fluorescence XAS	A2.1 - CZT detector and front-end design								
	A2.2 - Multichannel DAQ and signal-processing development								
	A2.3 - CZT calibration and performance characterization								
	A2.4 - Detector validation and operating-point selection						MS2		
WP3 - Instrument integration and commissioning	A3.1 - Procurement of source, optics and motion stages								
	A3.2 - Mechanical design and spectrometer assembly								
	A3.3 - Hardware, software and control-system integration								
	A3.4 - Instrument commissioning and verification							MS3	
WP4 - Ag reference samples and realistic sample demonstration	A4.1 - Ag foil calibration and reproducibility tests								
	A4.2 - XANES validation on Ag reference compounds								
	A4.3 - Fluorescence XAS demonstration on a realistic sample								
	A4.4 - Performance characterization and operating limits								MS4

M6 Design frozen
Simulations and specifications

M15 CZT qualified
Calibration dataset

M18 Instrument commissioned
Performance validated

M24 Ag demonstration
Validated XANES datasets

Team responsibilities and supporting collaborations

Simone Manti — 1.0 FTE
Project lead, simulations and measurements

Alessandro Scordo — 0.3 FTE
Crystal optics, ray tracing and integration

Massimiliano Bazzi — 0.3 FTE
Electronics and DAQ

Catalina Curceanu — 0.1 FTE
Scientific supervision and validation

Palermo and ENRICH COST
CZT development and simulation expertise

VOXES, KAONNIS and VIP
Infrastructure and detector experience

Due2Lab, Rigaku, Quantum Design
Letters of support

EuAPS and AQUA
Future transfer after laboratory validation

Resources for the integrated instrument

Year	Category	Item	Purpose/WP	Cost [€]
2027	Equipment	CZT Detector and Front-End	Fluorescence Detection - WP2	€30.000
2027	Equipment	Digitizer and DAQ	Signal Acquisition - WP2	€23.000
2027	Equipment	Motorized Stages and Controllers	Automated Positioning - WP3	€10.000
2027	Consumables	Ag Reference Targets	Calibration Tests - WP2/WP4	€2.000
2027	Services	Custom Mechanical Supports	Instrument Integration - WP3	€2.000
2027	Computing	Data Cloud CPU and Storage	End-to-End Simulations - WP1	€2.000
Total 2027				**€69.000**
2028	Equipment	Cooled X-ray Tube	Excitation Source - WP3	€10.000
2028	Equipment	X-ray Tube Power Supply	Source Operation - WP3	€7.000
2028	Travel	Oral Conference Travel	Results Dissemination - WP4	€4.000
2028	Equipment	Bent Crystal Optics	Energy Selection - WP3	€20.000
2028	Consumables	Additional Reference Targets	Multi-Edge Validation - WP4	€2.000
Total 2028				**€43.000**
Total				**€112.000**

TABLE I. Detailed financial request for the two-year PRISM project, broken down by year, cost category, item, and related Work Package.

112 k€

Requested budget
2027: 69 k€
2028: 43 k€

45 k€ estimated in kind
VOXES equipment
and infrastructure

Main risks and practical mitigation

Simulation accuracy	Benchmark against analytical calculations and VOXES measurements.
CZT optimization takes longer	Use the existing Si PIN for preliminary tests while CZT optimization continues.
Critical-component delays	Start alignment and integration with compatible VOXES components.
Low fluorescence rate or SNR	Optimize acceptance, scan steps and dwell times while preserving spectral quality.

PRISM: a validated laboratory fluorescence-XAS platform

Crystal optics and synchronized scanning

A commissioned instrument supported by source-to-detector simulations.

Room-temperature multichannel CZT detection

A calibrated detector, waveform readout and validated reconstruction methods.

Ag K-edge XANES as the validation benchmark

Quantitative reference data and a defined operating domain for realistic samples.

References

[1] Yamamoto. "Selection of Monochromator Crystal for Pb L- and Ag K-Edge XANES Using a Laboratory-Based Spectrometer." X-Ray Spectrometry (2026).

[2] Manti, Simone, and Alessandro Scordo. "Recent Developments of the VOXES Von Hamos X-Ray Spectrometer for Laboratory XES and XAS Studies." Engineering Innovations 19 (2026): 23-29.

[3] Scordo, A., et al. "VOXES: a high precision X-ray spectrometer for diffused sources with HAPG crystals in the 2–20 keV range." Journal of Instrumentation 13.04 (2018): C04002-C04002.

[4] Manti, Simone, et al. "Enhancing Performances of the VOXES Bragg Spectrometer for XES Investigations." Condensed Matter 9.1 (2024): 19.

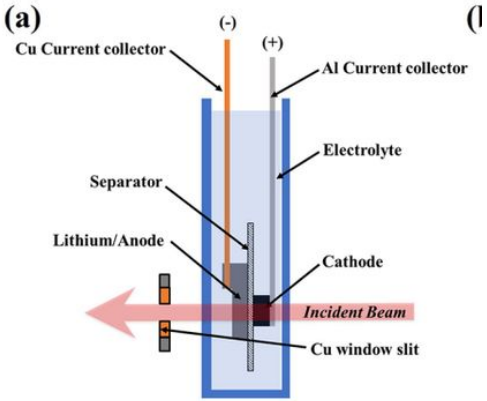
[5] Manti, Simone, et al. "Machine-learning-enhanced event selection for BEGe detectors in the VIP experiment." Journal of Instrumentation 21.01 (2026): P01036.

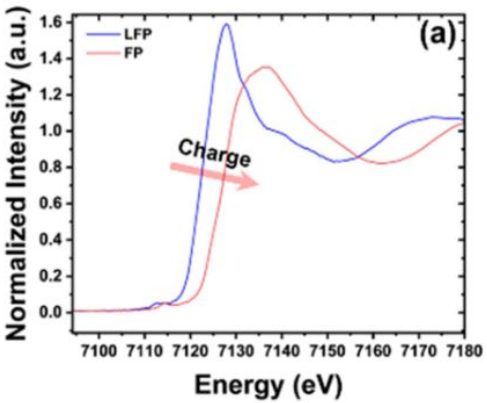
[6] Manti, Simone, et al. "Pulse-Shape Discrimination with Machine Learning for CZT Detectors at DAΦNE Beam Test Facility." Condensed Matter (2026).

SPARE

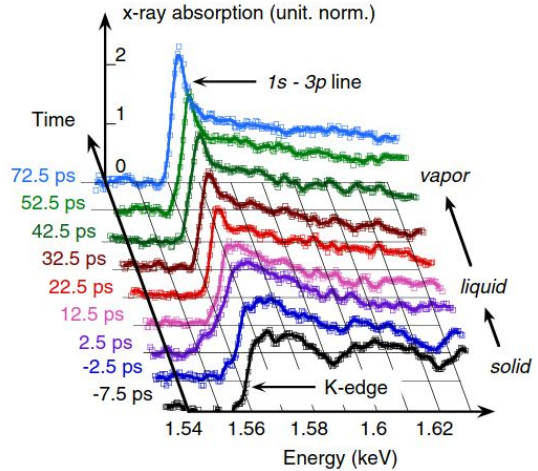
State-of-the-Art XAS: tracking materials during operation

- Element-specific sensitivity to **Oxidation State** and **Local Atomic Structure** enables studies of materials under operating conditions.
- **Battery Cycling:** Operando XAS tracks changes in the Fe oxidation state during charging.
- **Phase Transitions:** Ultrafast XANES follows the evolution from solid to liquid and vapor.





Krishnan et al. *ACS* 16.13 (2024)

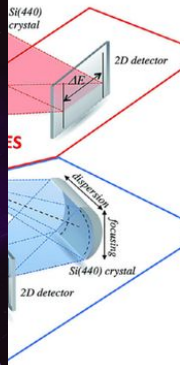


Dorchies, F., et al. *PRL* 107.24 (2011)

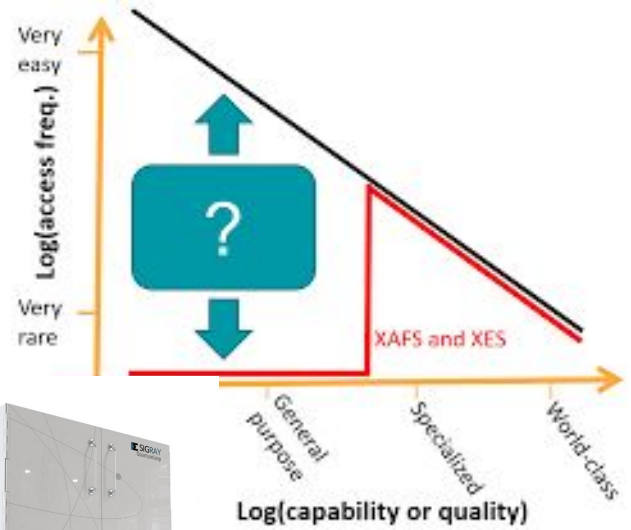



easyXAFS300



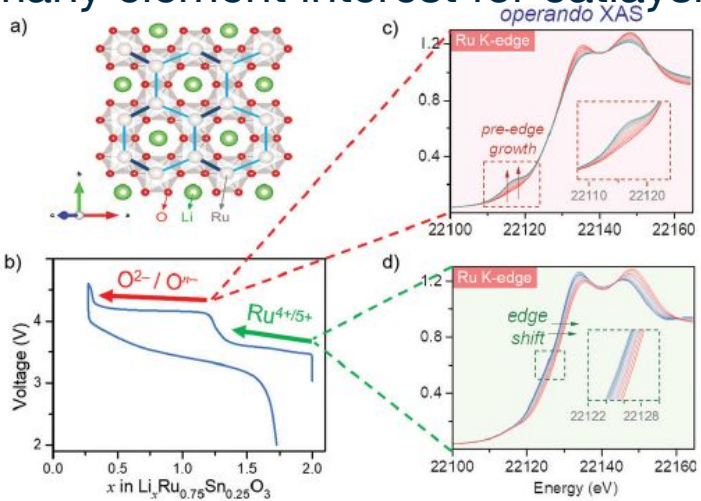

Si(440) crystal
2D detector
ΔE
dispersive
focusing
Si(440) crystal
2D detector

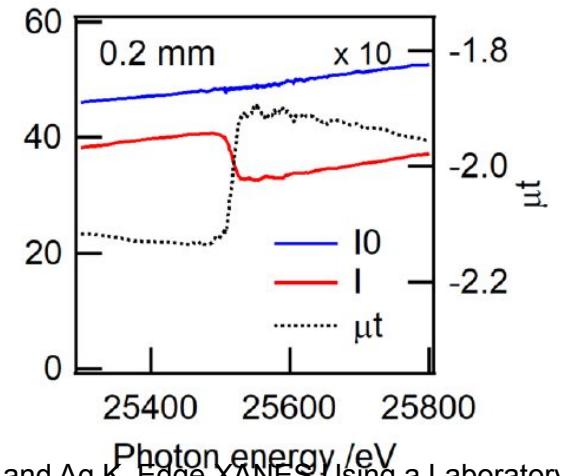



Very easy
Log(access freq.)
Very rare
?
XAFS and XES
General purpose
Specialized
World-class
Log(capability or quality)

Laboratory XAS above 20 keV: opportunities and challenges

- Current limitations are performing XAS for edge above 20 keV, where many element interest for catlaysi (Mo,Pd,Ag,Sn)



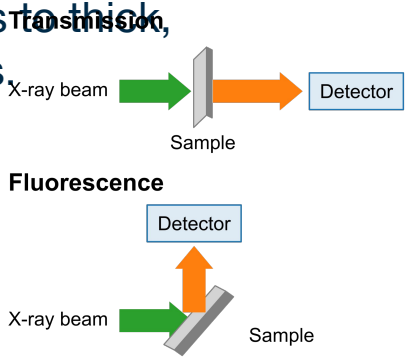


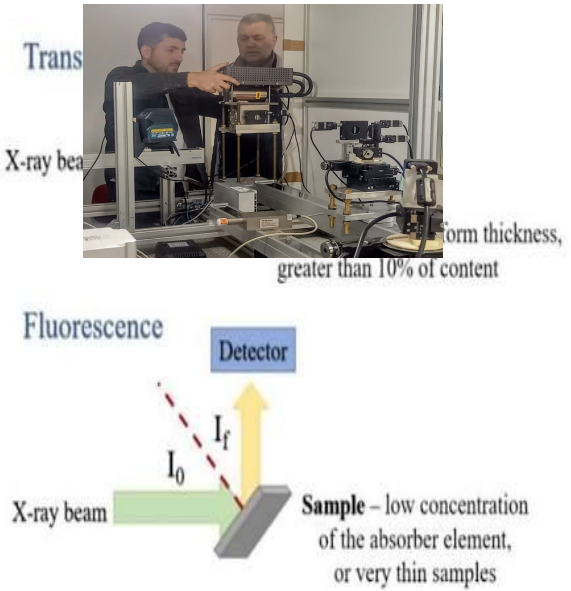
Yamamoto, Takashi. "Selection of Monochromator Crystal for Pb L-, Br K-, and Ag K-Edge XANES Using a Laboratory-Based Spectrometer." *X-Ray Spectrometry* (2026).

Why fluorescence detection mode for PRISM?

Transmission requires a suitable, homogeneous sample thickness.

Fluorescence extends access to thick, dilute or constrained samples.





Scan the incident energy.
Record fluorescence at each step.

Manti, Simone, and Alessandro Scordo. "Recent Developments of the VOXES Von Hamos X-Ray Spectrometer for Laboratory XES and XAS Studies." *Engineering Innovations* 19 (2026): 23-29.